Assessment - 1

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Course : Theory of Computation

Code : CSA1301

21. NFA to DFA

The given NFA has alphabet

$$\Sigma = \{a, b\}$$

1. NFA Transition table

state	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	$\emptyset$	$\{q_0, q_1\}$

$$A = \{q_0\}$$

for a:

$$\delta(A, a) = \delta(q_0, a) = \{q_0, q_1\} = B$$

for b:

$$\delta(A, b) = \delta(q_0, b) = \{q_2\} = C$$

State $B = \{q_0, q_1\}$

for a:

$$\delta(B, a) = \delta(q_0, a) \cup \delta(q_1, a)$$
$$= B$$

for b:

State $C = \{q_2\}$

for a:

$$\delta(C, a) = \emptyset = E$$

for b:

$$\delta(C, b) = \{q_0, q_1\} = B$$

State $D = \{q_1, q_2\}$

for a: $\delta(D, a) = \delta(q_1, a) \cup \delta(q_2, a)$

$= A$

for b: $\delta(D, b) = \delta(q_1, b) \cup \delta(q_0, b)$

$= \{q_1\} \cup \{q_0, q_1\} = B$

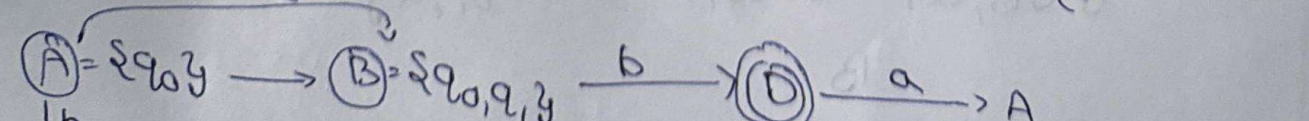
State $E = \emptyset$

for both inputs:

$$\delta(E, a) = E$$
$$\delta(E, b) = E$$

3. Equivalent DFA Transition Table

DFA State	a	b
$A = \{q_0\}$	B	C
$B = \{q_0, q_1\}$	B	D
$C = \{q_2\}$	E	B
$D = \{q_1, q_2\}$	A	B
$E = \emptyset$	E	E



$L = \{w \in \{a,b\}^* \mid |w| \geq 2,$ last 2 symbols are different$\}$

ab - accepted bb - rejected aba - accepted

$Q = \{q_0, q_1, q_2, q_3, q_f\}$

State	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\emptyset$	$\{q_f\}$
q_2	$\{q_f\}$	$\emptyset$
q_f	$\emptyset$	$\emptyset$

$A = \{q_0\}$

$B = \{q_0, q_1\}$

$C = \{q_0, q_2\}$

$D = \{q_0, q_1, q_f\}$

$E = \{q_0, q_2, q_f\}$

final states : D and E

DFA	a	b
A	B	C
B	B	D
C	E	C
D	B	D
E	E	C

Therefore, the DFA accepts exactly those strings

3. NFA accepting strings ending in 00 or 11

$\Sigma = \{0, 1\}$

$N = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3, q_f\}$

$\Sigma = \{0, 1\}$

$q_0 = q_0$

$F = q_f$

δ Transition function

State	0	1
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_f\}$	$\emptyset$
q_2	$\emptyset$	$\{q_f\}$
q_f	$\emptyset$	$\emptyset$

for 00:

$$q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_f$$

for 11:

$$q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_f$$

Successful path:

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_f$

00 or 11 NFA to DFA for strings containing 001

$L = \{w \in \{0,1\}^* \mid w \text{ contains } 001 \text{ as substring}\}$

Ex:-

001 - Accepted 000 - Rejected 0101 - Rejected

From q_0

If input is 0:

$q_0 \xrightarrow{0} q_1$

If input is 1:

$q_0 \xrightarrow{1} q_0$

$q_1 \xrightarrow{0} q_2$

$q_1 \xrightarrow{1} q_0$

$q_2 \xrightarrow{1} q_0$

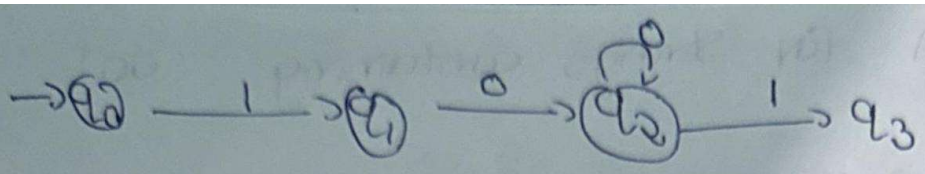
$q_2 \xrightarrow{0} q_2$

$q_3 \xrightarrow{0} q_3$

$q_3 \xrightarrow{1} q_3$

DFA Transition Table

State	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
q_3	q_3	q_3



10010 is accepted

5. Transition on 0

$A = \{q_0, q_1, q_2\}$

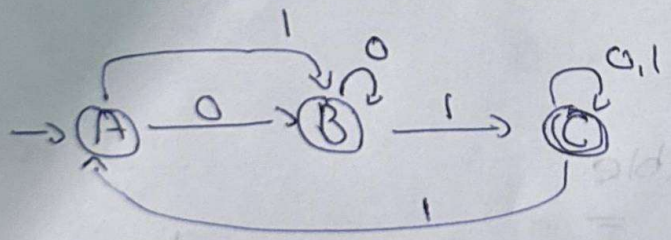
$q_1 \xrightarrow{0} q_1$ $\quad q_2 \xrightarrow{0} q_2$

Transition on 1

$q_0 \xrightarrow{1} q_0$ $\quad q_2 \xrightarrow{1} q_2$

Transition Table

DFA State	0	1
A $\{q_0, q_1, q_2\}$	B	A
$B = \{q_1, q_2\}$	B	C
$C = \{q_2\}$	C	C



ε-closure $(q_0\}$ $= \{q_0, q_1, q_2\}$

ε-closure (q_1) $= \{q_1, q_2\}$

ε-closure $= \{q_2\}$

final states: A, B, C

Thus, the equivalent DFA accepts the same languages as the given ε-NFA